

CAMBRIDGE IGCSE

10 Legacy matrices (removed from current syllabus)

0580 PAPER 4 - CALCULATOR

QUESTIONS + MARK SCHEME ANSWERS

2 complete questions

SUBTOPIC

Matrices (legacy content)

Each question is followed immediately by its official mark scheme answer.

9	(a)	$\begin{pmatrix} 2 & 13 \\ 1 & 14 \end{pmatrix}$	2	SC1 for one correct column or row	
	(b)	$\frac{1}{3}\begin{pmatrix} 3 & 2 \\ 0 & 1 \end{pmatrix}$ oe isw	2	B1 for $k\begin{pmatrix} 3 & 2 \\ 0 & 1 \end{pmatrix}$ oe for $k \neq 0$ or $\frac{1}{3}\begin{pmatrix} a & c \\ b & d \end{pmatrix}$	
	(c)	$\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$	3	B2 for two of $3 = u, 2u + 3v = 4u, 4 = 2 + v, u + 4v = 3 + 4v$ or B1 for one	
				or M1 for $\begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix} \begin{pmatrix} 0 & u \\ 1 & v \end{pmatrix} = \begin{pmatrix} 0 & u \\ 1 & v \end{pmatrix} \begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}$	
				B1 for $\begin{pmatrix} 3 & 2u+3v \\ 4 & u+4v \end{pmatrix}$ or $\begin{pmatrix} u & 4u \\ 2+v & 3+4v \end{pmatrix}$	
	(d)	12 nfww	2	M1 for $w \times 2 - 8 \times 3 [= 0]$ oe	

8	(a) (i)	Not possible	1	
	(ii)	$\begin{pmatrix} 4 & 0 \\ -2 & 10 \\ 6 & -8 \end{pmatrix}$ final answer	1	
	(iii)	$\begin{pmatrix} 14 & 35 \\ -8 & -20 \end{pmatrix}$ final answer	2	M1 for one correct column or row
	(iv)	(-6) final answer	2	M1 for $14 - 20$
	(v)	$\begin{pmatrix} -2 & 18 \\ -6 & 22 \end{pmatrix}$ final answer	2	M1 for one correct column or row

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